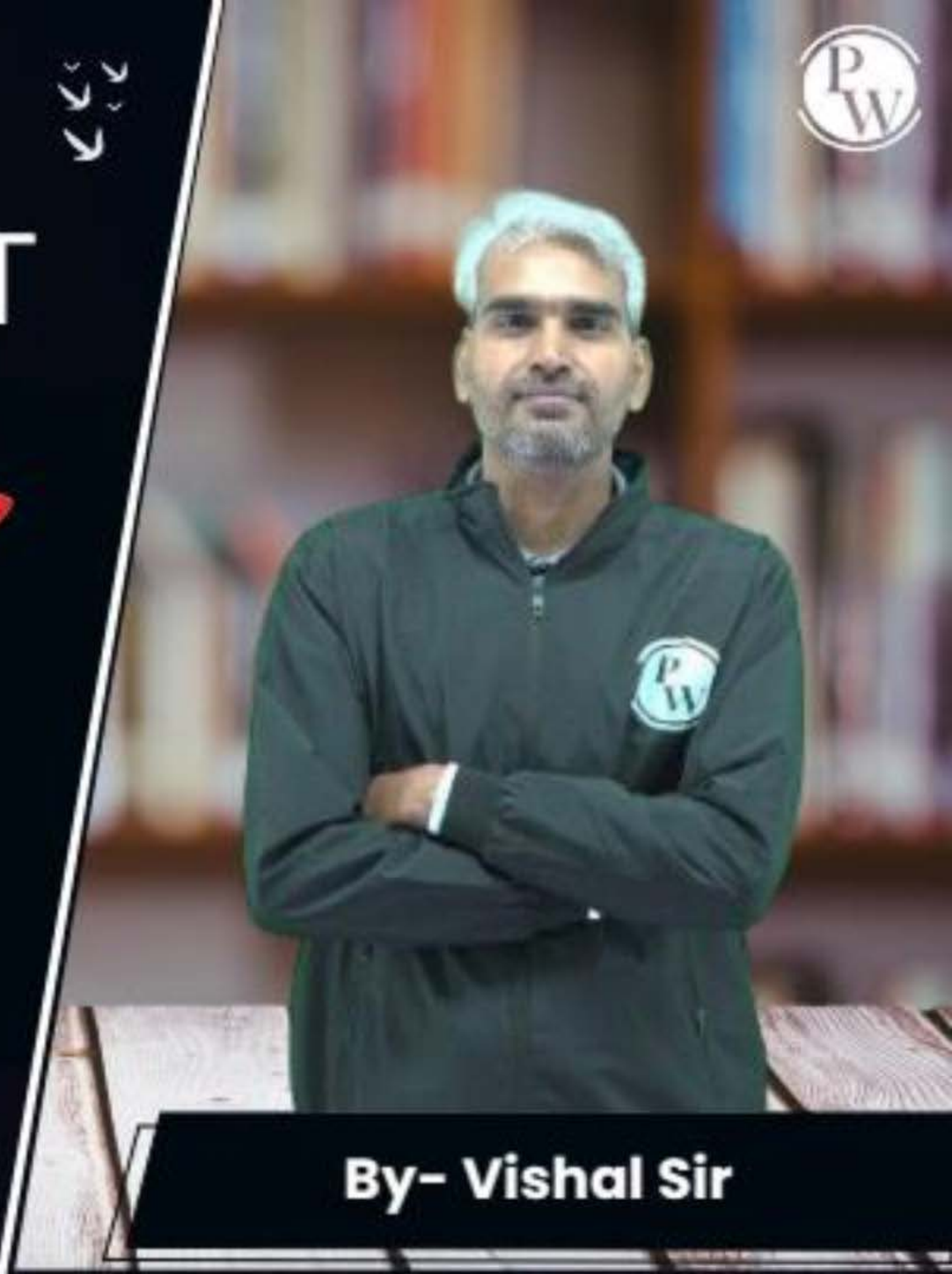
Computer Science & IT



Discrete Mathematics

Set Theory & Algebra

Lecture No. 11



By- Vishal Sir

Recap of Previous Lecture



- Topic: Equivalence relation
- Topic: Equivalence class
- Topic: Partition of a set
- Topic: Number of partitions of a set

Slide

Topics to be Covered



Topic — Number of partitions of a set

Topic — Number of equivalence relation on a set

Topic — Bell triangle

Slide



Topic : Partition of a set

- Partition of a "set A" is the set of non-Empty subsets of set A such that each element of set A appeares in Exactly one non-Empty subset of that set.

(or)

- Let A is any set and $A_1, A_2, A_3, \ldots A_k$ are non-Empty subsets of set A; then

$\{A_1, A_2, A_3, \ldots \ldots A_k\}$ is a partition of set A if and only if (i) $A_i \cap A_j = \phi$, $\forall i \neq j$ and (ii) $\bigcup_{i=1}^{k} A_i = A$

Slide

eg:- Let $A = \{1, 2, 3, 4, 5\}$
Which of the following is/are Partitions of set A.

✗ (a) $\{\{1,2,3\}, \{3,4\}, \{5\}\}$ — $\cap \neq \emptyset$ ∴ Not a Partition of A

✗ (b) $\{\{1,2\}, \{3\}, \{4\}\}$ — $\{1,2\} \cup \{3\} \cup \{4\} = \{1,2,3,4\} \neq A$ ∴ Not a partition, '5' is not included in any subset.

✗ (c) $\{\{1,2,3\}, \{4\}, \{5\}, \{\}\}$ — Empty subset of Set A is not allowed in Partition.

✓ (d) $\{\{1,2\}, \{3\}, \{4\}, \{5\}\}$

✓ (e) $\{\{1,2,3,4,5\}\}$ — is is partition of A w.r.t Equivalence Rel^n 'A×A'

✓ (f) $\{\{1,2\}, \{3\}, \{4,5\}\}$

* Find the number of partitions of "set A" when

(1) $|A| = 0$

(2) $|A| = 1$

(3) $|A| = 2$

(4) $|A| = 3$

(5) $|A| = 4$

① Let $A = \{\ \}$, i.e. $|A| = 0$

Partition of set A = { }

Partition of set A is a set such that it contains Non-Empty subsets of set A

But if 'A' itself is an Empty set, then there will be no non-empty subset of A, and hence no element will be present in the set w.r.t. Partition of set A

No element will be present.

Partition itself can be an Empty set, but Empty set can never be a member of a set corresponding to the Partition

i.e, **if $|A| = 0$ then number of partitions of set A = 1**

② Let $A = \{a\}$, i.e. $|A| = 1$

$|A| = 1 \Rightarrow \{1\}$
- Non-Zero, non-negative integer partition of '1'
- in the Partition there will only '1' subset of size = 1

Partition of set $A = \{\{a\}\}$
- it is the only partition possible w.r.t. set $A = \{a\}$

Partition of set A is a set
- such that elements will be non-empty subsets of set A
  - w.r.t. set $A = \{a\}$, there is only '1' non-empty subset and it is $\{a\}$ itself

i.e., **If $|A| = 1$ then number of partitions of set $A = 1$**

(3) Let $A = \{a, b\}$ i.e. $|A| = 2$

$|A| = 2 \Rightarrow$ { $1 + 1$ — Two subsets each of size = 1 (or) 2 — only one subset of size = 2 }

Non-zero non-negative integer partition of 2

Partitions of set $A = \{\{a, b\}\}$ and $\{\{a\}, \{b\}\}$

$\{\{a, b\}\} = \{\{b, a\}\}$

$\{\{a\}, \{b\}\} = \{\{b\}, \{a\}\}$

In this partition there is only one subset of size = 2

In this partition there are two subsets each of size = 1

i.e, If $|A| = 2$, then number of partitions of set $A = 2$

(4) Let $A = \{a, b, c\}$ i.e. $|A| = 3$

$|A| = 3 \Rightarrow 1+1+1$ — 3 subsets each of size = 1

$1+2 = 2+1$ — Two subsets one of size = 1 & other of size = 2

3 — only one subset of size = 3

Partitions of set $A = \{\{a,b,c\}\}$ & $\{\{a\},\{b,c\}\}$ & $\{\{a\},\{b\},\{c\}\}$

&

$\{\{b\},\{a,c\}\}$

&

$\{\{c\},\{a,b\}\}$

i.e, **If $|A| = 3$, then Number of partitions of set $A = 5$**

④ Let $A = \{a, b, c\}$ i.e. $|A| = 3$

$|A| \Rightarrow 3 \Rightarrow 1+1+1$

$1+2$

3

Partitions of set $A = \{\{_\},\{_\},\{_\}\}$ or $\{\{_\},\{_,_\}\}$ or $\{\{_,_,_\}\}$

$\{a\}\ \{c\}\ \{b\}$
$\{a\},\{b\}\ \{c\}$
$\{b\},\{a\}\ \{c\}$
$\{b\},\{c\}\ \{a\}$
$\{c\},\{a\}\ \{b\}$
$\{c\},\{b\}\ \{a\}$

$\frac{{}^3C_1 * {}^2C_1 * {}^1C_1}{3!}$ or ${}^3C_1 * {}^2C_2$ or ${}^3C_3$

$= 1$ $= 3 * 1 = 3$ $= 1$

i.e, If $|A| = 3$, then Number of partitions of set $A = 1 + 3 + 1 = 5$

$|A| = 1 \Rightarrow 1$

$|A| = 2 \Rightarrow 1 + 1$

2

$|A| = 3 \Rightarrow 1 + 1 + 1$

$1 + 2 = 2 + 1$

3

$|A| = 4 \Rightarrow 1 + 1 + 1 + 1$

$1 + 1 + 2$

$1 + 3$

$2 + 2$

4

$|A| = 5 \Rightarrow$

$1+1+1+1+1$

$1+1+1+2$

$1+1+3$

$1+2+2$

$1+4$

$2+3$

5

$|A| = 6 \Rightarrow$

$1+1+1+1+1+1$

$1+1+1+1+2$

$1+1+1+3$

$1+1+2+2$

$1+1+4$

$1+2+3$

$1+5$

$2+2+2$

$2+4$

$3+3$

6

⑤ Let $A = \{a, b, c, d\}$, i.e. $|A| = 4$, then find no. of Partitions of set A

$|A| = 4 \Rightarrow 1+1+1+1 = \dfrac{{}^4C_1 * {}^3C_1 * {}^2C_1 * {}^1C_1}{4!} = 1$

$1+1+2 = \dfrac{{}^4C_1 * {}^3C_1 * {}^2C_2}{2!} = 6$

$1+3 = {}^4C_1 * {}^3C_3 = 4$

$2+2 = \dfrac{{}^4C_2 * {}^2C_2}{2!} = 3$

$4 = {}^4C_4 = 1$

if $|A| = 4$ then number of Partitions of Set $A = 1+6+4+3+1 = 15$

Today's Topic

H.W.

⑥ Let $A = \{a, b, c, d, e\}$ i.e. $|A| = 5$,
Find the total no. of Partitions possible of set A

$|A| = 5 \Rightarrow$

$1+1+1+1+1 = \frac{{}^5C_1 * {}^4C_1 * {}^3C_1 * {}^2C_1 * {}^1C_1}{5!} = 1$

$1+1+1+2 = \frac{{}^5C_1 * {}^4C_1 * {}^3C_1 * {}^2C_2}{3!} = 10$

$1+1+3 = \frac{{}^5C_1 * {}^4C_1 * {}^3C_3}{2!} = 10$

$1+2+2 = \frac{{}^5C_1 * {}^4C_2 * {}^2C_2}{2!} = 15$

$1+4 = {}^5C_1 * {}^4C_4 = 5$

$2+3 = {}^5C_2 * {}^3C_3 = 10$

$5 = {}^5C_5 = 1$

Total no. of Partitions possible

$= 1+10+10+15+5+10+1$

$= \boxed{52}$ Ans

$|A| = 10$

$2 + 2 + 3 + 3$

$$\frac{{}^{10}C_2 * {}^{8}C_2}{2!} * \frac{{}^{6}C_3 * {}^{3}C_3}{2!} = 4$$

$\{a,b\} * \{f,g\}$

$a, b, c, d, e, f, g, h, i, j$

$S_1 \quad S_3 \quad S_2 \quad S_4$

$\{a,b\}, \{c,d,e\}, \{f,g\}, \{h,i,j\}$

$\{f,g\} \quad \{c,d,e\} \quad \{a,b\} \quad \{h,i,j\}$

$\{f,g\} \quad \{h,i,j\} \quad \{a,b\}, \quad \{c,d,e\}$

$\{a,b\} \quad \{h,i,j\}, \quad \{f,g\}, \quad \{c,d,e\}$

H.W.

$|A| = 6 \Rightarrow$

$1+1+1+1+1+1 = 1$

$1+1+1+1+2 = \frac{6\times5\times4\times3}{4!} \times {}^2C_2 = 15$

$1+1+1+3 = \frac{6\times5\times4}{3!} \times {}^3C_3 = 20$

$1+1+2+2 = \frac{6\times5}{2!} \times \frac{{}^4C_2 \times {}^2C_2}{2!} = 45$

$1+1+4 = \frac{6\times5}{2!} \times {}^4C_4 = 15$

$1+2+3 = 6 \times {}^5C_2 \times {}^3C_3 = 60$

$1+5 = 6 \times {}^5C_5 = 6$

$2+2+2 = \frac{{}^6C_2 \times {}^4C_2 \times {}^2C_2}{3!} = 15$

$2+4 = {}^6C_2 \times {}^4C_4 = 15$

$3+3 = \frac{{}^6C_2 \times {}^3C_3}{2!} = 10$

$6 = 1$

∴ Total no. of Partitions Possible $= 203$

(7) Let $A = \{a, b, c, d, e, f\}$

i.e. $|A| = 6$,

Find total no. of partitions possible of Set A.

* Find the number of partitions of "set A" when

(1) $|A| = 0 \Rightarrow 1$

(2) $|A| = 1 \Rightarrow 1$

(3) $|A| = 2 \Rightarrow 2$

(4) $|A| = 3 \Rightarrow 5$

(5) $|A| = 4 \Rightarrow 15$

(6) $|A| = 5 \Rightarrow 52$

(7) $|A| = 6 \Rightarrow 203$

NOTE

* If we know the Equivalence relation on set A, then we can obtain the Partition of set A w.r.t. given Equivalence relation.

Partition of the set can be obtained by defining a set that contains all distinct Equivalence classes w.r.t. given Equivalence relation

eg:- let $A = \{1, 2, 3, 4, 5, 6\}$

and $R = \{(1,1), (2,2), (3,3), (4,4), (5,5), (6,6), (1,2), (2,1), (1,3), (3,1), (2,3), (3,2), (4,6), (6,4)\}$ is an Equivalence relation on set A.

Find the Partition of set A, w.r.t. Equivalence relation R

$[1] = \{1, 2, 3\}$

$[2] = \{1, 2, 3\}$

$[3] = \{1, 2, 3\}$

$[4] = \{4, 6\}$

$[5] = \{5\}$

$[6] = \{4, 6\}$

$\therefore$ Partition of set A w.r.t. Equivalence relation R is

$= \{\{1, 2, 3\}, \{4, 6\}, \{5\}\}$

NOTE

* If we know the partition of set A, then we can identify the Equivalence relation on set A corresponding to which the given partition of set A is created.

  * If we perform the self-cross-product of all the subsets of set A present in the partition, and if we union the result of all those self-cross products, then we get the desired Equivalence relation

eg: Let $A = \{1, 2, 3, 4, 5\}$
and R is an Equivalence relation on set A.

$R = \{(1,1), (2,2), (3,3), (4,4), (5,5), (1,2), (2,1), (4,5), (5,4)\}$

We know partition of set A w.r.t. Equivalence relation R is $= \{\{1,2\}, \{3\}, \{4,5\}\}$

Equivalence Rel^n R $= \{1,2\} \times \{1,2\} \cup \{3\} \times \{3\} \cup \{4,5\} \times \{4,5\}$

$= \{(1,1), (1,2), (2,1), (2,2)\} \cup \{(3,3)\} \cup \{(4,4), (4,5), (5,4), (5,5)\}$

$= \{(1,1), (2,2), (3,3), (4,4), (5,5), (1,2), (2,1), (4,5), (5,4)\}$

eg:- let $A = \{1, 2, 3, 4, 5, 6\}$

and $R = \{(1,1), (2,2), (3,3), (4,4), (5,5), (6,6), (1,2), (2,1), (1,3), (3,1), (2,3), (3,2), (4,6), (6,4)\}$ is an Equivalence relation on set A.

We know,

Partition of set A w.r.t. relation R is $= \{\{1,2,3\}, \{4,6\}, \{5\}\}$

Equivalence relation $R = \{1,2,3\} \times \{1,2,3\} \cup \{4,6\} \times \{4,6\} \cup \{5\} \times \{5\}$

NOTE

For Every partition of set A, there is unique Equivalence relation, and with respect to Every Equivalence relation on set A there is unique partition of set A.



Topic : Number of equivalence relation on a set

Number of Equivalence relation on set A = Number of Partitions of set A.

Slide

* Find the number of Equivalence relations on "set A" when

(1) $|A| = 0$, then number of Equivalence relation on set A = 1

(2) $|A| = 1$, then number of Equivalence relation on set A = 1

(3) $|A| = 2$, then number of Equivalence relation on set A = 2

(4) $|A| = 3$, then number of Equivalence relation on set A = 5

(5) $|A| = 4$, then number of Equivalence relation on set A = 15

(6) $|A| = 5$, then number of Equivalence relation on set A = 52

(7) $|A| = 6$ then number of Equivalence relation on set A = 203



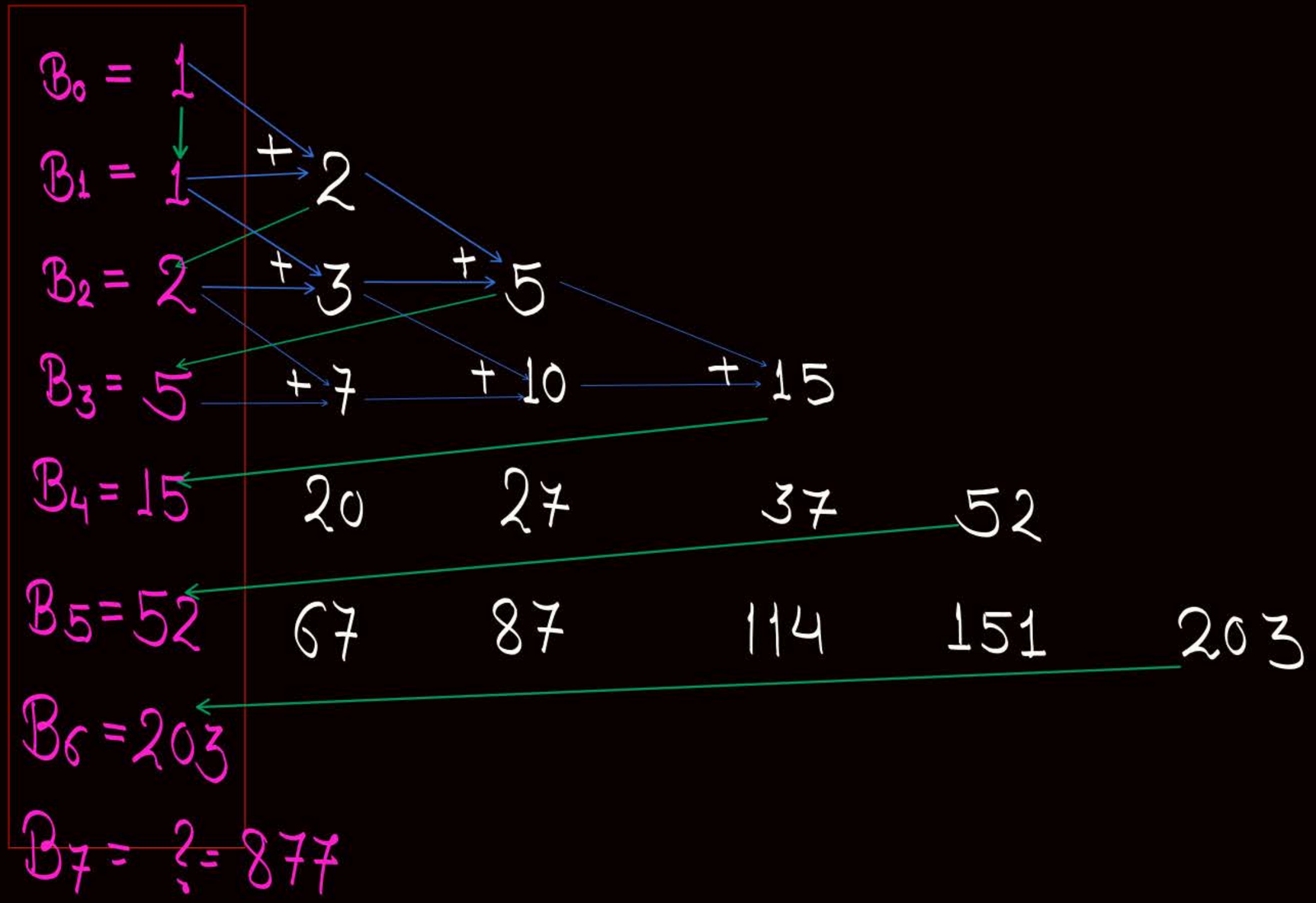
Topic : Bell Number

* Bell number B_n will represent the number of partions of a set A of size = n.

  ↳ i.e; B_n will represent the number of Equivalence relation on a set A of cardinality = n

Slide

Bell Triangle :-



Q: Let $A = \{a, b, c, d, e\}$ i.e. $|A| = 5$

How many equivalence relations are possible on set A, such that cardinality of each equivalence relation is exactly '3'.

No. of order pairs in Equivalence relation w.r.t Partition of Corresponding type

$|A| = 5 \Rightarrow$

$1+1+1+1+1$
$(1\times1)+(1\times1)+(1\times1)+(1\times1)+(1\times1) = 5$

$1+1+1+2$
$(1\times1)+(1\times1)+(1\times1)+(2\times2) = 7$

$1+1+3$
$(1\times1)+(1\times1)+(3\times3) \longrightarrow = 11$

$1+2+2$
$(1\times1)+(2\times2)+(2\times2) \longrightarrow = 9$

← No. of order pairs will be '9' in Equivalence rel^n, w.r.t. Partitions in which there are two subsets of size = 2 & one subset of size = 1

and no. of such Partitions

$= {}^5C_1 * \frac{{}^4C_2 * {}^2C_2}{2!} = 15$

i.e. '15' Equivalence rel^n on set A s.t. Cardinality of Equivalence rel^n 9

Q: Let $A = \{a, b, c, d, e, f\}$ i.e. $|A| = 6$

How many equivalence relations are possible on set A, such that cardinality of each equivalence relation is exactly '12'.

$|A| = 6 \Rightarrow$

No. of order pairs in equivalence relⁿ

$1+1+1+1+1+1 \quad 1+1+1+1+1+1 = 6$

$1+1+1+1+2 \quad 1+1+1+1+4 = 8$

$1+1+1+3 \quad 1+1+1+9 = 12$

$1+1+2+2 \quad 1+1+4+4 = 10$

$1+1+4 \quad 1+1+16 = 18$

$1+2+3 \quad 1+4+9 = 14$

$1+5 \quad 1+25 = 26$

$2+2+2 \quad 4+4+4 = 12$

$2+4 \quad 4+16 = 20$

$3+3 \quad 9+9 = 18$

$6 \quad 36 = 36$

Total no. of Equivalence relⁿ Possible on set A s.t. Cardinality of each Equivalence relⁿ is Exactly = 12

$= \left(\frac{{}^6C_1 \times {}^5C_1 \times {}^4C_1 \times {}^3C_3}{3!}\right) + \left(\frac{{}^6C_2 \times {}^4C_2 \times {}^2C_2}{3!}\right)$

$= 20 + 15$

$= 35$



2 mins Summary

Topic Number of partitions of a set

Topic Number of equivalence relation on a set

Topic Bell triangle

Slide

Next Class:

"Partial orders"

THANK - YOU

Slide